

HVAC Opta I/O Pin Chart

5 configs · one unit per column

Per-unit pin comparison — facility/RTU use D1608E; split unit (res/com) = base Opta only
MooreviewOptaMqttSt 2.3.81 · Generated 2026-08-14

Overview

Five configs: facility dual-unit columns (one unit each), RTU, and res/com split-unit base-Opta templates. Dual boards stack unit 2 on higher base/expansion pins.

- Facility + RTU: Opta + D1608E expansion. Split unit (res/com): base Opta analog I1–I8 only (no D1608E).
- All motor CT: single-phase with start cap (1P+cap, MCSA). Pan leak: analog mV on /ahu-env (never digital DI).
- Facility IoT-Link: 2 Optas (dual AHU + dual cond) + IoT-Link + EZ Meter.
- Split unit templates: hvac-split-unit-ahu · hvac-split-unit-cond (1 Opta each).

Chart 1 — Signal map (function !' pin)

Function	Fac. AHU	Fac. cond	RTU	Split unit AHU	Split unit cond
Supply / blower start CT	I1	—	I1	I1	—
Supply / blower run CT	I2	—	I2	I2	—
Condenser fan start CT	—	I1	I3	—	I1
Condenser fan run CT	—	I2	I4	—	I2
Compressor start CT	—	I5	I5	—	I3
Compressor run CT	—	I6	I6	—	I4
Supply air NTC	I5	—	X1_IRAW3	I3	—
Return air NTC	I6	—	X1_IRAW4	I4	—
Refrigerant low-side NTC	—	X1_IRAW1	X1_IRAW1	—	I5
Refrigerant high-side NTC	—	X1_IRAW2	X1_IRAW2	—	I6
Pan leak (analog mV)	X1_IRAW1	—	X1_IRAW5	I5	—
Commissioning	/ct-cal + /ahu-env	/mcsa	/mcsa + /ahu-env	/mcsa + /ahu-env	/mcsa
MooreVIEW driver	opta_ahu	opta_cond	opta_rtu	opta_split_ahu	opta_split_cond
Default MQTT device ID	hvac_ahu_01	hvac_cond_01	hvac_rtu_01	hvac_split_ahu_01	hvac_split_cond_01

Chart 2 — Base Opta pin allocation (I1–I8)

Direct mapping of each base analog input for one logical unit. Dual facility boards offset unit 2 to I3/I4/I7/I8 (+ expansion).

Pin	Fac. AHU	Fac. cond	RTU	Split unit AHU	Split unit cond
I1	Fan start CT	Fan start CT	Blower start CT	Blower start CT	Fan start CT
I2	Fan run CT	Fan run CT	Blower run CT	Blower run CT	Fan run CT
I3	—	—	Fan start CT	Supply NTC	Comp start CT
I4	—	—	Fan run CT	Return NTC	Comp run CT
I5	Supply NTC	Comp start CT	Comp start CT	Pan leak (mV)	Lo-side NTC
I6	Return NTC	Comp run CT	Comp run CT	—	Hi-side NTC
I7	—	—	—	—	—
I8	—	—	—	—	—

Chart 3 — D1608E expansion (X1_IRAW)

Facility AHU / cond / RTU only. Split unit configs use base Opta only (no expansion row assignments).

Pin	Fac. AHU	Fac. cond	RTU	Split unit AHU	Split unit cond
X1_IRAW1	Pan leak (mV)	Lo-side NTC	Lo-side NTC	—	—
X1_IRAW2	—	Hi-side NTC	Hi-side NTC	—	—
X1_IRAW3	—	—	Supply NTC	—	—
X1_IRAW4	—	—	Return NTC	—	—
X1_IRAW5	—	—	Pan leak (mV)	—	—

Chart 4 — Dual-unit board stacking

Board	Unit 1 pins	Unit 2 pins
Dual AHU — 1× Opta + D1608E	I1/I2 CT · I5/I6 NTC · X1_IRAW1 leak	I3/I4 CT · I7/I8 NTC · X1_IRAW2 leak
Dual cond — 1× Opta + D1608E	I1/I2 fan CT · I5/I6 comp CT · X1_IRAW1/2 NTC	I3/I4 fan CT · I7/I8 comp CT · X1_IRAW3/4 NTC
RTU — 1× Opta + D1608E	I1–I6 CT + X1_IRAW1–5 (single rooftop)	—
Split unit AHU — res/com · base Opta	I1/I2 blower CT · I3/I4 NTC · I5 leak	—
Split unit cond — res/com · base Opta	I1–I4 fan/comp CT · I5/I6 NTC	—

Projects & generators

- facility-hvac-iot-link — dual AHU + dual cond + EZ Meter (npm run build:facility-hvac-iot-link)
- hvac-rtu-opta — RTU template (npm run build:hvac-rtu-opta)
- hvac-split-unit-ahu — res/com split unit AHU (npm run build:hvac-split-unit-ahu)
- hvac-split-unit-cond — res/com split unit condenser (npm run build:hvac-split-unit-cond)
- Chart data: scripts/hvac-io-map/chart-data.js · PDF: npm run build:hvac-io-map-chart-pdf

Legend

Term	Meaning
CT	Current transformer input (0–1 V on Opta analog)
NTC	10K thermistor — 24 V divider
X1_IRAWn	D1608E expansion analog input n
Pan leak	Spot leak sensor — analog mV, wet threshold on /ahu-env
/mcsa	MCSA monitor web UI — motor CT + refrigerant NTC
/ahu-env	AHU env cal — supply/return NTC + pan leak mV